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DRISHTI Innovation Challenge | Problem Statement-71

AURORA

Automated Unified Reasoning and Optimization Architecture

CONCEPT NOTE

Submitted to Mazagon Dock Shipbuilders Limited (MDL)
In response to the DRISHTI Innovation Challenge

Field	Details
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Challenge	Problem Statement-71, Design Optimization, Compliance Validation & CAD-BOM Synchronization
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AURORA functions as an engineering decision-support system for design optimization, compliance validation, and CAD-BOM synchronization. Final engineering approval authority remains with designated MDL personnel. All outputs generated by the system are advisory in nature and are intended to support, not replace, formal engineering review and approval workflows.

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Executive Summary

Naval ship design and validation workflows currently rely on extensive manual coordination between CAD models, Bill of Materials (BOM), classification rule libraries, drawings, and engineering review teams. These workflows are fragmented across disconnected engineering systems and heavily dependent on institutional expertise, resulting in prolonged validation cycles, elevated rework risk, configuration inconsistencies, and delayed engineering decision-making.

AURORA (Automated Unified Reasoning and Optimization Architecture) is proposed as an integrated, on-premise engineering decision-support platform for Mazagon Dock Shipbuilders Limited (MDL) under DRISHTI Problem Statement-71. The system unifies CAD data, BOM information, classification rules, simulation references, and engineering dependencies into a single traceable operational environment capable of supporting complex naval engineering programmes.

The platform enables automated CAD-BOM discrepancy detection, multi-society compliance validation, change impact propagation analysis, drawing revision synchronization, and engineering optimisation support while maintaining complete auditability and mandatory human oversight. By enabling change-driven validation instead of repeated full-system review cycles, AURORA is designed to significantly reduce manual cross-verification effort and accelerate engineering workflows.

Expected operational benefits include reduction of validation timelines from weeks to hours, improved configuration consistency, earlier identification of downstream engineering impacts, enhanced traceability of design decisions, and long-term preservation of institutional engineering knowledge across vessel programmes.

AURORA is designed specifically for defence-grade operational environments and supports fully air-gapped, on-premise deployment with no dependency on external cloud infrastructure. The platform follows a deterministic-first architecture in which compliance-critical workflows remain rule-based, traceable, and auditable. Selective intelligent optimisation and pattern-detection modules are applied where appropriate, while all engineering approvals, certification actions, and production release decisions remain fully under authorised MDL personnel.

The architecture is modular, scalable, and integration-ready for existing CAD, PLM, ERP, and engineering ecosystems, supporting long-term engineering assurance and lifecycle management requirements.

Operational Area	Existing Workflow	With AURORA
CAD-BOM Validation	Manual cross-verification	Automated synchronization
Compliance Validation	Repeated full-system review	Change-triggered validation
Change Impact Assessment	SME dependent	Dependency-driven propagation
Drawing Revision Tracking	Manual coordination	Automated synchronization
Engineering Traceability	Distributed records	Unified audit trail
Deployment Environment	Multiple disconnected tools	Unified on-premise platform

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Preface: Decision-Support Role & Scope Boundary

Scope Boundary

AURORA is designed as an engineering decision-support platform for design optimization, compliance validation, CAD-BOM synchronization, and change impact analysis within naval ship design workflows. The platform is intended to assist engineering teams by reducing manual cross-verification effort, improving traceability, and accelerating validation processes.

AURORA does not replace detailed CAD authoring tools, structural simulation software, formal certification workflows, or authorised engineering approval processes. Final certification, manufacturing release, and classification society approvals remain fully under authorised MDL engineering and programme management personnel.

The platform specifically does not perform:

- Autonomous CAD redesign or parametric geometry generation
- Native real-time structural simulation (FEA/CFD) computation; simulation outputs are ingested and evaluated externally
- Formal classification society certification issuance
- Autonomous manufacturing release or production-floor instruction generation

Deterministic-First Design Philosophy

AURORA follows a deterministic-first architecture for all compliance-critical engineering workflows. Core capabilities including compliance validation, CAD-BOM reconciliation, dependency impact propagation, and drawing synchronization operate through rule-based, traceable, and auditable logic designed to produce consistent and explainable outputs.

Selective intelligent optimisation and pattern-detection modules are applied where operationally beneficial, including assisted CAD-BOM confidence scoring and engineering optimisation support. These modules operate strictly in an advisory capacity and remain fully subject to engineer review and approval.

In scenarios where optional intelligent modules are unavailable or insufficient training data exists, AURORA continues operating in deterministic mode using rule-based validation and threshold-driven matching with no degradation to core compliance validation or CAD-BOM synchronization functionality.

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1. Features & Functionalities

AURORA (Automated Unified Reasoning and Optimization Architecture) is an integrated, on-premise engineering decision-support platform designed for naval ship design workflows at Mazagon Dock Shipbuilders Limited (MDL). The system addresses DRISHTI Problem Statement-71 by unifying design data, automating compliance validation workflows, and synchronising CAD and BOM information within a secure air-gapped operational environment.

AURORA is designed to transform manual and error-prone cross-verification activities into structured, traceable, and accelerated engineering workflows while supporting improved configuration consistency, dependency visibility, and decision traceability across complex naval programmes.

The platform follows a modular multi-layer architecture comprising input ingestion, unified component registry, dependency knowledge graph, engineering intelligence engines, drawing synchronisation, enterprise integration services, and a unified decision-support dashboard. This architecture enables phased deployment, incremental integration with existing engineering ecosystems, and future scalability across vessel programmes.

Core System Features

- **Unified Component Registry :** Integrates CAD, BOM, simulation references, and engineering metadata into a single traceable environment.
- **Multi-Format CAD Ingestion :** Supports ingestion and normalization of STEP, IGES, JT, CATIA, NX, and SolidWorks design data.
- **Dependency Knowledge Graph :** Maps inter-system engineering relationships for structured dependency analysis and impact propagation.
- **CAD-BOM Synchronization :** Detects discrepancies, revision mismatches, and configuration inconsistencies across engineering datasets.
- **Multi-Society Compliance Validation :** Evaluates engineering designs against IRS, DNV, Lloyd's Register, BV, ABS, and related rule frameworks.
- **Change Impact Analysis :** Identifies downstream engineering, structural, and routing impacts resulting from design modifications.
- **Drawing Revision Synchronization :** Tracks drawing status, revision history, and propagation of approved engineering changes.
- **Engineering Optimisation Support :** Assists engineers with trade-off analysis across weight, cost, compliance, and design constraints.
- **Classification Workflow Management :** Supports submission package generation, query tracking, and re-submission coordination for classification societies.
- **Decision Intelligence Dashboard :** Provides a unified interface for compliance review, discrepancy analysis, impact assessment, and operational monitoring.
- **Audit Trail & Role-Based Access Control :** Maintains immutable activity records with controlled access for engineering, review, and administrative roles.
- **Air-Gapped Defence Deployment :** Operates fully on-premise with no dependency on external cloud infrastructure or internet connectivity.
- **Enterprise Integration Layer:** Supports phased integration with existing PLM, ERP, CAD, and engineering management ecosystems.

1.1 Unified Component Registry Core

The Unified Component Registry serves as AURORA's central engineering data environment, maintaining a structured digital representation of ship components, assemblies, and associated engineering metadata. Each component is linked across CAD models, BOM records, compliance rules, and revision history to establish a consistent and traceable engineering reference framework.

For every registered component, the system maintains:

- Unique component identifier linked to CAD assembly hierarchy and BOM records
- Geometric and physical properties including dimensions, volume, surface area, and centre-of-gravity data
- Material and engineering attributes including grade, density, and structural properties
- Associated BOM metadata including quantity, supplier reference, and revision state
- Linked compliance and classification rule references
- Version-controlled change history with traceable revision records

AURORA is designed to support ingestion of major naval and industrial CAD formats including STEP, IGES, JT, CATIA, NX, and SolidWorks, along with structured BOM data formats such as XML and CSV. Final deployment compatibility will be aligned with the CAD and engineering ecosystems currently operational within MDL during the integration assessment phase.

By maintaining a unified engineering reference environment, the Unified Component Registry reduces inconsistencies between design, BOM, and compliance datasets while significantly reducing the manual cross-verification effort traditionally required during naval design validation workflows.

1.2 Dependency Knowledge Graph Engine

AURORA represents ship systems and engineering interdependencies through a structured Dependency Knowledge Graph that models relationships between components, subsystems, engineering rules, and supporting infrastructure. The graph acts as the platform's central dependency reasoning framework, enabling structured analysis of how design modifications propagate across interconnected ship systems.

The Dependency Knowledge Graph captures multiple categories of engineering relationships including:

- Structural support dependencies between load-bearing components
- Electrical and signal distribution dependencies
- Thermal proximity and environmental interaction relationships
- Routing dependencies for cable trays, piping, and supporting infrastructure
- Mechanical fastening and assembly relationships

- Classification and compliance rule associations linked to affected components

By maintaining these dependency relationships in a unified graph structure, AURORA enables automated downstream impact analysis, dependency-aware compliance evaluation, and improved visibility into engineering change propagation across complex naval programmes.

The graph framework is designed to support large-scale vessel assemblies containing thousands of interconnected components. Graph completeness and dependency accuracy are validated during deployment and integration phases through engineering review workflows, with unresolved or isolated components flagged for Subject Matter Expert (SME) assessment prior to operational use.

1.3 CAD Parsing & Normalisation Engine

The CAD Parsing & Normalisation Engine enables ingestion and standardisation of multi-format engineering design data into the AURORA environment. The engine is designed to process CAD assemblies from multiple industrial and naval design ecosystems and convert them into a structured representation suitable for downstream compliance validation, dependency analysis, and CAD-BOM synchronization workflows.

Core capabilities of the engine include:

- Multi-format CAD ingestion and geometry conversion
- Assembly hierarchy extraction and component identification
- Detection and correction of incomplete or inconsistent geometry data
- Extraction of geometric and physical engineering properties
- Standardisation of component metadata for downstream processing
- Geometry-based change detection for revision and impact analysis workflows

The engine supports major industrial CAD formats including STEP, IGES, JT, CATIA, NX, and SolidWorks. Final deployment compatibility and workflow alignment will be configured according to the CAD ecosystems operational within MDL during the integration assessment phase.

By transforming heterogeneous CAD data into a unified and structured engineering representation, the engine enables consistent downstream processing across compliance validation, dependency analysis, BOM synchronization, and drawing revision workflows while significantly reducing manual preprocessing effort.

1.4 CAD-BOM Synchronisation Engine

The CAD-BOM Synchronisation Engine establishes correspondence between CAD assembly structures and Bill of Materials (BOM) records to ensure configuration consistency across engineering datasets. The engine is designed to identify missing components, revision mismatches, material inconsistencies, duplicate records, and mass-related discrepancies that traditionally require extensive manual cross-verification during naval engineering workflows.

The synchronisation workflow evaluates multiple engineering validation signals simultaneously to determine alignment between CAD assemblies and BOM entries.

Validation Signal	Operational Purpose
Geometry Correspondence	Confirms alignment between CAD geometry and BOM-linked component references
Part Metadata Similarity	Identifies consistency between component names, descriptions, and engineering metadata
Mass & Property Validation	Detects discrepancies in mass, material properties, and physical attributes
Assembly Relationship Validation	Verifies assembly hierarchy, positional consistency, and component placement relationships
Revision Alignment	Detects outdated, duplicate, or inconsistent revision states across engineering datasets

The engine generates structured discrepancy reports highlighting:

- Missing or unmatched components
- Material inconsistencies
- Mass and weight deviations
- Duplicate records
- Revision mismatches between CAD and BOM datasets

AURORA supports both deterministic rule-based matching and assisted confidence-based evaluation workflows. In environments where validated historical datasets are limited or unavailable, the platform continues operating using deterministic engineering validation logic with no dependency on intelligent matching modules for core synchronization functionality.

By automating CAD-BOM reconciliation and discrepancy detection, the engine significantly reduces manual validation effort while improving engineering consistency, traceability, and revision control across complex naval ship design programmes.

1.5 Multi-Society Rule Compliance Engine

The Multi-Society Rule Compliance Engine performs deterministic compliance evaluation of engineering parameters against applicable classification society rule frameworks. The engine is designed to provide consistent, traceable, and auditable compliance assessment across ship components, assemblies, and associated engineering datasets.

For identical engineering inputs, the platform produces consistent validation outputs with every finding traceable to the corresponding rule reference and evaluation context. The engine is designed to support rule frameworks from major classification societies including:

- Indian Register of Shipping (IRS)
- Lloyd's Register (LR)
- DNV
- Bureau Veritas (BV)
- American Bureau of Shipping (ABS)

The compliance framework supports multiple operational validation modes and rule management capabilities.

Capability	Operational Purpose
Baseline Compliance Scan	Performs full-system compliance evaluation across registered components and associated engineering rules
Change-Triggered Validation	Re-evaluates only rules affected by approved design or drawing modifications to reduce unnecessary review cycles
Multi-Society Conflict Identification	Flags conflicting rule requirements between classification frameworks for engineering review and resolution
Rule Catalog Version Management	Maintains historical rule references and revision traceability for audit and retrospective validation workflows
Air-Gapped Rule Update Support	Supports secure offline rule catalog updates for defence-grade operational environments

Compliance findings are categorised according to engineering severity and operational impact to support prioritised review workflows and traceable engineering resolution management.

The engine is designed to operate fully within an air-gapped on-premise environment and supports integration with engineering review, approval, and audit workflows. Final rule framework configuration, severity classification standards, and applicable society mappings will be aligned with MDL operational requirements during deployment and validation phases.

1.6 Change Impact Analysis & Propagation Engine

The Change Impact Analysis & Propagation Engine evaluates how engineering modifications propagate across interconnected ship systems and dependent components. When a design, material, routing, or configuration change is introduced, the engine performs structured dependency analysis using the Dependency Knowledge Graph to identify downstream systems, assemblies, and compliance relationships potentially affected by the modification.

The impact propagation workflow supports:

- Identification of directly and indirectly affected components

- Dependency-aware analysis across structural, electrical, thermal, routing, and mechanical relationships
- Prioritisation of high-impact engineering changes for review
- Detection of downstream compliance and configuration implications
- Targeted re-validation of affected engineering domains
- Traceable propagation paths for engineering audit and review workflows

The propagation engine applies weighted dependency evaluation to distinguish critical engineering impacts from low-relevance relationships, ensuring that review workflows remain focused on operationally significant changes. Propagation depth and sensitivity parameters are configurable according to project complexity, vessel type, and engineering review requirements.

By enabling structured change propagation analysis, AURORA significantly reduces manual dependency tracing effort while improving engineering visibility, configuration consistency, and downstream impact awareness during naval design modification workflows.

1.7 Design Optimisation Engine (NSGA-II)

AURORA includes an Engineering Optimisation Engine designed to support evaluation of design trade-offs across parameters such as weight, cost, compliance, and engineering constraints. The engine assists engineers in exploring alternative design configurations while maintaining alignment with applicable classification and operational requirements.

The optimisation framework supports:

- Multi-parameter engineering trade-off analysis
- Evaluation of alternative design configurations
- Comparison of weight, cost, and compliance impacts
- Prioritisation of feasible engineering alternatives
- Structured side-by-side comparison of candidate solutions
- Parameter sensitivity analysis for constrained design scenarios

Where sufficient engineering parameters and bounded datasets are available, the platform may apply assisted multi-objective optimisation workflows to generate ranked design alternatives for engineering review. In scenarios where optimisation conditions are limited or insufficient engineering data exists, the system instead provides structured parameter sensitivity and dependency analysis to support informed engineering decision-making.

The optimisation framework is designed strictly as an advisory engineering support capability. All optimisation outputs remain fully subject to engineering validation, operational review, and authorised approval workflows prior to implementation.

Future enhancement pathways may include integration of simulation-assisted optimisation workflows and accelerated surrogate modelling techniques as larger validated engineering datasets become available during later deployment phases.

1.8 Drawing Synchronisation & Revision Engine

The Drawing Synchronisation & Revision Engine manages controlled propagation of approved engineering changes into associated production and engineering drawings. The engine is designed to improve drawing consistency, revision traceability, and coordination between CAD models, BOM datasets, and released engineering documentation.

The engine supports:

- Automated synchronisation of approved design revisions into engineering drawings
- Revision tracking and controlled update workflows
- Detection of outdated or inconsistent drawing states
- Drawing status management across review and approval stages
- Traceable linkage between engineering changes and associated drawing revisions
- Structured revision audit history and change logging

Supported workflows include handling of industrial drawing formats such as DXF, DWG, and PDF within controlled engineering documentation environments.

Capability	Operational Purpose
Automated Revision Synchronisation	Propagates approved engineering changes into associated drawings and documentation
Drawing State Monitoring	Detects outdated, inconsistent, or unsynchronised drawing revisions
Revision Traceability	Maintains structured history of engineering revisions and associated change records
Production Readiness Statusing	Tracks drawings through review, approval, and production-release stages
PLM-Integrated Drawing Management	Supports integration with enterprise drawing vaults and engineering document management systems

The engine is designed to support integration with existing PLM and engineering document management ecosystems operational within MDL. Final workflow configuration, drawing approval states, and integration pathways will be aligned with MDL engineering and production processes during deployment and validation phases.

By automating drawing synchronisation and revision tracking workflows, AURORA reduces manual documentation coordination effort while improving engineering traceability, configuration consistency, and production-readiness visibility across naval ship design programmes.

1.9 Classification Society Submission Package

AURORA includes a structured Classification Society Submission Package framework designed to support engineering submission, review, and response workflows across applicable naval and maritime classification authorities. The framework consolidates compliance findings, engineering documentation, revision history, and supporting records into a traceable submission environment aligned with existing engineering approval processes.

The submission workflow supports generation of:

- Rule-wise compliance evaluation summaries with traceable engineering references
- Design calculation and engineering assessment summaries
- Drawing revision and configuration traceability records
- Structured discrepancy and resolution reports
- Submission-ready engineering documentation packages for review workflows

The framework additionally supports structured handling of classification society review queries and engineering clarification cycles.

Capability	Operational Purpose
Pre-Submission Validation	Performs final consistency and compliance checks prior to engineering submission
Submission Package Generation	Consolidates compliance findings, drawing references, and engineering summaries into structured submission workflows
Society Query Management	Tracks engineering queries, clarification requests, and review comments linked to affected components and revisions
Targeted Re-Evaluation Workflow	Re-validates only impacted engineering domains following approved changes or review observations
Revision & Re-Submission Tracking	Maintains traceable history of submissions, revisions, resolutions, and re-submission cycles

The platform is designed to support the complete engineering review lifecycle:
Pre-Submission Review → Submission → Query Resolution → Re-Validation → Re-Submission

By structuring and automating classification submission workflows, AURORA reduces manual coordination effort, improves traceability of engineering responses, and accelerates review and approval cycles across complex naval engineering programmes.

1.10 AURORA Decision Intelligence Dashboard

The AURORA Decision Intelligence Dashboard provides engineers, programme managers, and authorised reviewers with a unified operational interface across all major system workflows and engineering outputs. The dashboard is designed to consolidate compliance validation, CAD-BOM synchronization, impact analysis, drawing revision management, and engineering review activities into a single traceable decision-support environment.

The dashboard supports centralised monitoring and management of:

- Engineering system health and operational status
- Design change and impact analysis workflows
- CAD-BOM discrepancy review and resolution
- Compliance validation and rule evaluation findings
- Drawing revision and production-readiness tracking
- Classification society submission and query management
- Engineering optimisation and trade-off analysis outputs
- Audit trail review and engineering activity traceability

The dashboard framework supports role-based operational access for engineering, review, approval, and administrative workflows while maintaining full traceability of user actions, engineering decisions, and approval events.

Dashboard Capability	Operational Purpose
System Monitoring & Status	Provides visibility into engineering workflow health, synchronisation status, and operational alerts
Impact Analysis Workspace	Supports review of engineering change propagation and affected systems
CAD-BOM Review Interface	Enables identification and resolution of configuration inconsistencies and discrepancies
Compliance Validation Workspace	Consolidates rule evaluation findings and engineering compliance review workflows

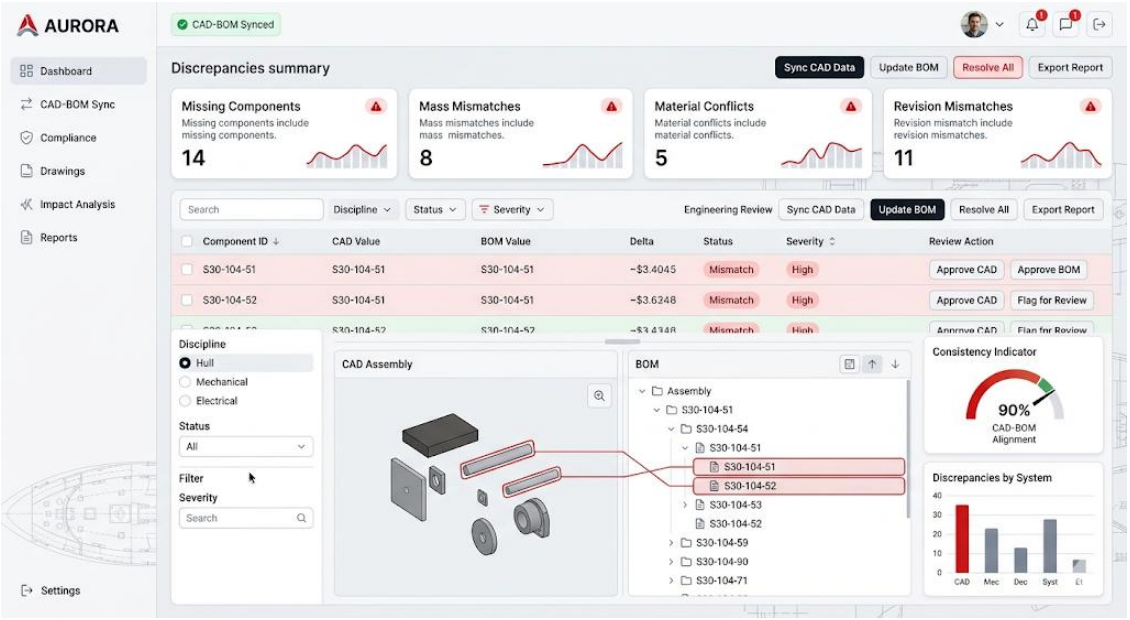
Drawing & Revision Management	Tracks drawing states, revision history, and production-readiness workflows
Classification Workflow Management	Supports submission tracking, query handling, and re-submission coordination
Engineering Optimisation Workspace	Provides comparative evaluation of alternative engineering configurations
Audit & Traceability Interface	Maintains searchable records of engineering actions, decisions, and approval history

The dashboard is designed for fully on-premise deployment within air-gapped defence environments and supports phased integration with existing engineering, PLM, and enterprise management ecosystems operational within MDL.

By consolidating engineering intelligence, compliance workflows, and revision management into a unified operational interface, AURORA improves engineering visibility, reduces coordination overhead, and enhances traceability across complex naval ship design programmes.

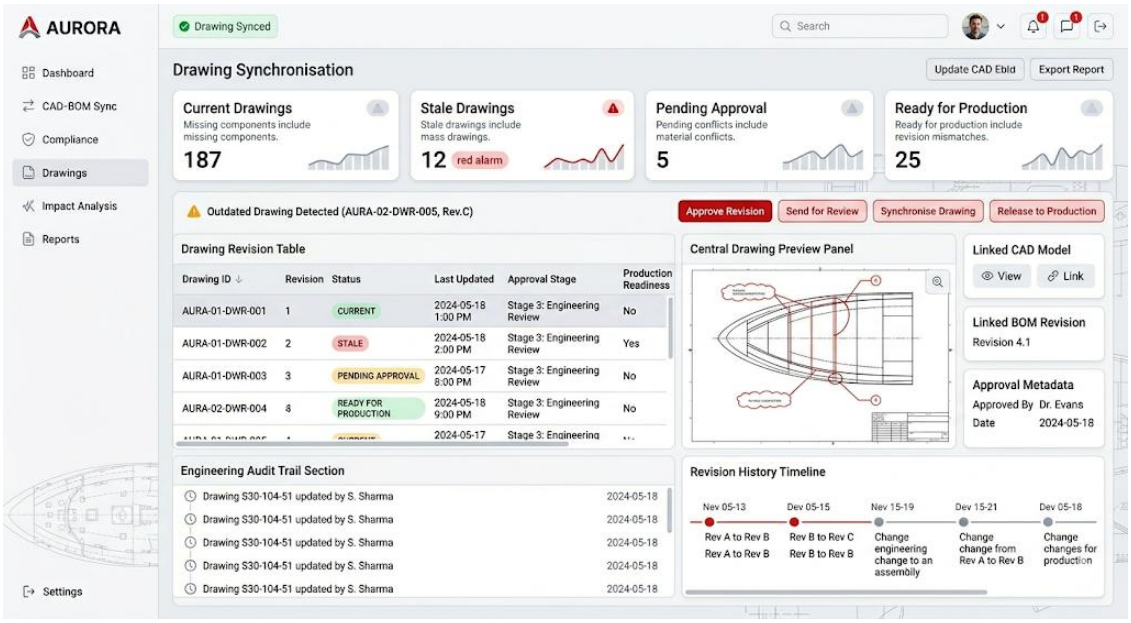
Note: The below dashboard images are representative conceptual illustrations intended solely for demonstration purposes. Final interface design and workflow configuration will be adapted to MDL operational requirements during implementation.

Sample Dashboard View

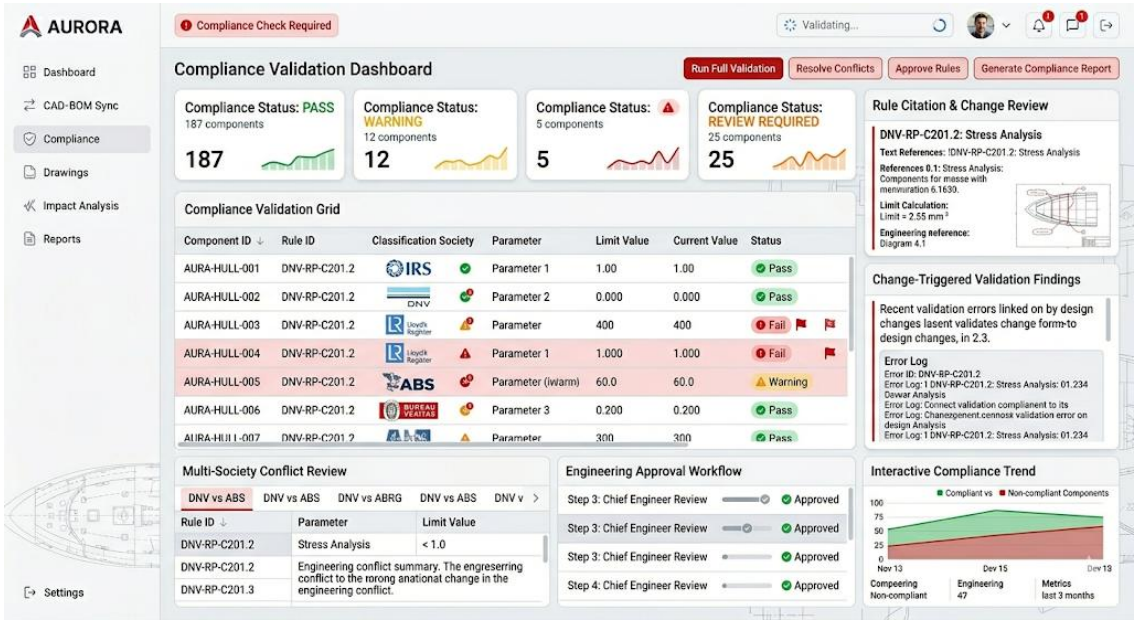


Cad-BOM Sync Dashboard

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Drawing Synchronization Dashboard



Compliance Validation Dashboard

2. Usage, Usability & Operational Constraints

2.1 Intended Users and Roles

The AURORA platform supports role-based operational access to ensure controlled engineering workflows, traceable approvals, and secure separation of responsibilities across design, validation, review, and administrative functions.

Role	Operational Responsibilities
Viewer (Programme Manager / Observer)	Read-only access to all dashboard panels, impact reports, and audit trail
Analyst (Design / Systems Engineer)	Submit change requests, trigger all engine analyses, flag discrepancies, view all outputs
Approver (Lead Engineer / Programme Director)	All Analyst permissions + approve/reject drawings, sign change records with digital approval, manage re-submission packages
Admin (IT / System Administrator)	Full access + user management, RBAC configuration, rule catalog editing, audit trail administration

Final role definitions and workflow permissions will be aligned with MDL operational and security requirements during deployment.

2.2 Typical Operational Workflow

A standard design change assessment proceeds through the following staged workflow:

Stage	Operational Workflow
Stage 1 : CAD & BOM Ingestion	Updated CAD assemblies and BOM datasets are ingested into the platform and linked to the Unified Component Registry for structured processing and traceability.
Stage 2 : CAD-BOM Synchronisation	The system performs automated reconciliation between CAD assemblies and BOM records, identifying discrepancies and inconsistencies for engineering review.
Stage 3 : Change Impact Analysis	Dependency-aware analysis identifies affected components, subsystems, and downstream engineering domains impacted by the proposed modification.

Stage 4 : Compliance Validation	Applicable classification and engineering rules associated with impacted systems are re-evaluated and consolidated into structured compliance findings.
Stage 5 : Engineering Optimisation (Optional)	Where applicable, the platform provides comparative engineering trade-off analysis and alternative configuration evaluation for engineer consideration.
Stage 6 : Drawing Synchronisation	Associated engineering drawings and revision-controlled documentation are updated and synchronised with approved design modifications.
Stage 7 : Classification Submission Workflow	Consolidated engineering findings, compliance summaries, and supporting documentation are prepared for classification review and submission workflows.
Stage 8 : Approval & Audit Archiving	Approved engineering actions, revision history, compliance records, and workflow decisions are archived within the platform audit and traceability framework.

The workflow is designed to maintain complete engineering traceability while reducing manual coordination effort across CAD validation, compliance review, drawing management, and classification submission activities.

Human review and approval checkpoints remain integrated throughout all operational stages to ensure engineering governance, auditability, and controlled release management.

2.3 Usability Provisions

- Role-based interface hides unnecessary controls, reducing operator error and cognitive load
- Change-triggered findings displayed separately from baseline findings to prevent information overload
- All compliance findings cite specific rule ID, actual value, and allowable limit in plain engineering units, not abstract scores
- Interactive dependency graph visualisation with colour-coded impact severity (CRITICAL = Red, HIGH = Orange, MEDIUM = Amber, LOW = Green), filterable by ship system or deck
- XGBoost match confidence scores displayed alongside each CAD-BOM match, enabling transparent engineer oversight
- Mandatory engineer review checkpoint before any approval action : no autonomous approvals
- One-click PDF report export auto-named with change_id and timestamp for programme review boards
- All overrides recorded with mandatory justification text in immutable audit trail

- Dashboard served fully on-premise with no browser dependency on external resources; no internet **required**

2.4 Operational Constraints

The following operational constraints and deployment considerations are applicable to the proposed AURORA platform architecture and implementation approach:

- **Air-Gapped Deployment Environment** : AURORA is designed for fully on-premise deployment within defence-grade operational environments with no dependency on external cloud infrastructure or continuous internet connectivity after deployment.
- **Enterprise Authentication Integration** : The platform is designed to integrate with existing enterprise authentication and access-control ecosystems operational within MDL, including directory-based identity management and role-based access control frameworks.
- **Deterministic Core Functionality** : Core platform capabilities including compliance validation, CAD-BOM synchronization, dependency analysis, and drawing traceability are designed to operate from initial deployment without dependency on historical training datasets or intelligent optimisation modules.
- **CAD Ecosystem Compatibility** : Final compatibility and integration pathways will depend on the CAD formats, workflows, and engineering ecosystems operational within MDL. Format assessment and interoperability validation will be conducted during initial deployment and integration phases.
- **Phased Rule Framework Expansion** : Initial deployment phases may prioritise high-criticality and high-frequency engineering rule domains, with progressive expansion of rule coverage and validation workflows during subsequent deployment stages.
- **PLM & Enterprise Integration Dependency** : Full integration with existing PLM, ERP, and engineering management systems will be aligned with MDL interface specifications, operational workflows, and available integration APIs during implementation phases.
- **Engineering Validation Dependency** : All compliance rules, workflow configurations, and operational validation criteria remain subject to Subject Matter Expert (SME) review and approval prior to operational activation within production environments.

Reference Deployment Hardware Configuration :
The proposed benchmark and performance metrics within this concept note are based on a reference on-premise deployment configuration comprising:

- Multi-core enterprise-grade CPU infrastructure
- Minimum 64–128 GB ECC RAM
- Dedicated GPU acceleration for optional optimisation and analytical workflows
- High-speed NVMe storage for CAD and graph-processing workloads

3. Performance Parameters & Metrics

The following performance metrics represent indicative operational targets based on the proposed reference deployment configuration, representative engineering datasets, and phased deployment assumptions described within this concept note. Actual operational performance may vary depending on vessel complexity, engineering workflow scale, integration environment, and deployment configuration within MDL operational infrastructure.

3.1 CAD-BOM Synchronisation Performance Targets

Metric	Indicative Target
CAD-BOM Match Precision	Target ≥ 90% against representative pilot validation datasets
CAD-BOM Match Recall	Target ≥ 90% against representative pilot validation datasets
False Positive Reduction	Confidence-based review workflows designed to minimise incorrect automated matches
Critical Compliance False Negatives	System objective is elimination of critical compliance-related false negatives
Deterministic Cold-Start Matching	Designed to support operational synchronization workflows without dependency on historical training datasets
Mass & Property Consistency Validation	Significant deviations automatically flagged for engineering review

3.2 Compliance Validation Performance

Metric	Indicative Target
SME Validation Agreement	Target ≥ 85% agreement with Subject Matter Expert (SME) review workflows
Critical Rule Violation Detection	System objective is identification of all critical compliance violations within validated rule scope
Change-Triggered Validation Coverage	All rules associated with approved engineering modifications re-evaluated during delta validation workflows
Rule Traceability Coverage	All compliance findings linked to traceable rule references and engineering evaluation context
Multi-Society Conflict Identification	Contradictory or overlapping rule conditions automatically flagged for engineering review

3.3 System Performance & Throughput

Metric	Indicative Target
CAD Assembly Ingestion	Target benchmark based on representative large-scale vessel assemblies under reference deployment configuration
Change Impact Analysis	Target response time of a few seconds for standard subsystem propagation workflows
Compliance Delta Validation	Target completion within minutes for isolated engineering modifications
Full Baseline Compliance Evaluation	Designed to support full-vessel validation workflows within operationally practical engineering review timelines
Dashboard Response Time	Designed for responsive interaction during standard engineering review workflows
Concurrent User Support	Multi-user engineering workflow support under representative deployment configuration
Engineering Optimisation Workflow	Accelerated execution under GPU-enabled deployment environments where applicable

3.4 Validation & Acceptance Benchmarks

Pilot validation activities are proposed during phased deployment and user acceptance testing workflows using representative historical engineering scenarios provided by MDL.

Indicative acceptance objectives include:

- CAD-BOM synchronization precision and recall targets validated against representative historical engineering datasets
- Compliance validation agreement benchmarking against Subject Matter Expert (SME) review outcomes
- Verification of critical compliance violation detection within validated rule scope
- Validation of dependency-aware impact propagation across known engineering change scenarios
- Verification of drawing synchronization and revision traceability workflows
- User Acceptance Testing (UAT) and operational review sign-off by authorised MDL engineering personnel

3.5 Knowledge Graph Completeness Metric

- Target: ≥ 95% of BOM line items with ≥ 1 dependency edge before go-live (Phase 1 gate)
- Isolated nodes (0 edges) flagged and surfaced to SME for manual review
- Edge weight calibration validated via SME workshop review in Phase 0
- Graph completeness dashboard panel displays real-time coverage metrics throughout deployment

3.6 Intelligent Assistance & Analytical Performance Metrics

The following indicative metrics are proposed for evaluation of AURORA's intelligent assistance and analytical support workflows under representative deployment and validation conditions:

Metric	Indicative Target
Assisted CAD-BOM Validation Agreement	≥ 90% agreement with SME-reviewed engineering validation outcomes
Intelligent Discrepancy Prioritisation Accuracy	≥ 85% alignment between prioritised discrepancies and SME-assessed criticality
Engineering Optimisation Recommendation Relevance	≥ 80% positive evaluation from participating engineering reviewers during pilot validation
Explainability & Traceability Coverage	100% of intelligent analytical outputs linked to traceable engineering context, source references, or supporting validation logic
Unsupported / Non-Traceable Analytical Outputs (Hallucination)	System objective is elimination of non-traceable or unsupported analytical recommendations within validated operational scope
Human Review Enforcement	100% of optimisation, compliance, and approval workflows require authorised engineer review prior to operational acceptance
Deterministic Fallback Availability	100% continuity of core engineering workflows without dependency on intelligent analytical modules
Critical Compliance False Negative Objective	System objective is elimination of critical compliance-related false negatives within validated operational rule scope
Recommendation Transparency	100% of analytical recommendations explicitly labelled as advisory with engineer override capability

Optional Analytical Verification Layer

AURORA's intelligent assistance workflows may optionally integrate with a Hallucination Firewall (Veritect) made by Arcaisys which acts as a response-verification and analytical consistency layer, designed to reduce unsupported or non-traceable analytical outputs during AI-assisted engineering workflows.

The verification layer operates by:

- Cross-validating analytical outputs against deterministic engineering constraints and structured rule references
- Enforcing traceability linkage between generated recommendations and validated engineering context
- Flagging low-confidence or insufficiently supported analytical responses for mandatory engineer review
- Preventing unsupported outputs from bypassing established engineering approval workflows

The verification framework is designed as an additional assurance mechanism supplementing AURORA's deterministic-first and human-in-the-loop operational philosophy.

4. Integration & Verification to Target Platform

4.1 Integration Architecture

All engine interactions are orchestrated by a FastAPI + Uvicorn async backend exposing a REST API gateway. Integration follows a hub-and-spoke model with AURORA's API Gateway as the central hub.

Integration Target	Protocol	Data Exchanged	Notes
PLM : Teamcenter / Windchill	REST Adapter Stubs	BOM, drawing metadata, revision history, approval status	Adapter stubs provided; full integration activated on client environment access during Phase 0
ERP : SAP PM / Oracle	REST / OData	Cost rates, supplier data, BOM sync	ERP-side configuration required; bi-directional BOM synchronisation
CAD Systems : CATIA, NX, SolidWorks	File Watcher / Webhook	CAD models, assembly hierarchy, geometry hashes	On-save hooks trigger CAD ingestion pipeline; webhook endpoint configurable per format
Async Task Queue : Celery + Redis	Internal	Job status, engine outputs, error logs	All long-running jobs (CAD parsing, NSGA-II, ML inference) dispatched via Celery; status polling via REST

4.2 Deployment Architecture

The platform architecture supports phased deployment, scalable infrastructure expansion, and integration with existing engineering and enterprise ecosystems operational within MDL.

The deployment framework is designed to support:

- Air-gapped operation with no dependency on continuous external connectivity
- Modular deployment across development, pilot, and production environments
- Scalable multi-service orchestration for enterprise-grade operational workloads
- Secure on-premise hosting of engineering, compliance, and audit datasets

- Controlled role-based access management and authentication integration
- High-availability and maintainable infrastructure deployment workflows

Component	Technology / Specification
API Backend	FastAPI + Uvicorn (ASGI); Python 3.10+
Frontend Dashboard	React + TypeScript; served as static bundle from Nginx on-premise
Primary Database	PostgreSQL 15 (containerised, ACID-compliant) for Unified Component Registry and audit trail
Graph Database	Neo4j (containerised) for Dependency Knowledge Graph
Object Storage	MinIO (containerised, S3-compatible; all telemetry disabled) for CAD files, simulation results, model artefacts
Task Queue Broker	Redis 7.x (containerised, no external connections) + Celery
Authentication	On-premise LDAP/Active Directory; JWT tokens for API session management
Container Registry (K8s)	Docker Registry v2 (containerised) : no external registry access required

4.3 Air-Gap Deployment Package

AURORA is designed for deployment within fully air-gapped defence operational environments with no dependency on continuous internet or external cloud connectivity after installation. The deployment framework supports secure offline installation, controlled update management, and long-term maintainability within isolated engineering infrastructure.

The air-gapped deployment package includes:

- Pre-packaged containerised application environments for offline installation and deployment
- Offline-bundled software dependencies and engineering processing libraries
- Secure local deployment of databases, workflow services, and engineering storage systems
- Offline installation and update workflows requiring no external package repository access after deployment
- Version-controlled engineering datasets, rule catalogs, and analytical model artefacts
- Deployment, recovery, and operational maintenance documentation for on-premise administration

Optional advanced optimisation and analytical workflows may utilise dedicated GPU acceleration within the reference deployment environment. However, all core deterministic engineering capabilities including compliance validation, CAD-BOM synchronization, dependency analysis, and drawing management remain fully operational without GPU dependency.

Rule catalog and platform updates are designed for secure offline delivery through controlled package validation and cryptographic integrity verification workflows aligned with defence-grade operational security requirements.

4.4 Phase 0 Integration Assessment

Prior to production integration activities, AURORA includes a structured Phase 0 assessment and deployment planning stage intended to align the platform with MDL's operational engineering environment, infrastructure constraints, and enterprise integration requirements.

The Phase 0 assessment includes:

- Evaluation of CAD formats, engineering workflows, and interoperability requirements within MDL operational ecosystems
- Assessment of PLM, ERP, and engineering management interface specifications for integration planning
- Review of network topology, deployment zones, authentication pathways, and internal communication requirements
- Alignment of BOM structures, engineering metadata, and data schemas with AURORA's unified component registry framework
- Assessment of historical engineering datasets and availability of validated reference data for optional advanced analytical workflows
- Identification of deployment constraints, infrastructure dependencies, and operational security considerations
- Joint review and validation of workflow requirements with engineering and IT stakeholders

The assessment phase is intended to minimise integration risk, identify compatibility constraints early, and establish a validated deployment roadmap prior to operational rollout activities.

5. Test Plans & Procedures for End-User Trials

5.1 Testing Strategy Overview

End-user trials for AURORA are intended to validate operational suitability, engineering workflow integration, compliance traceability, and usability within representative MDL deployment environments. Trial activities are designed to confirm that the platform operates reliably within air-gapped naval engineering workflows while supporting structured decision-making, engineering review, and traceable approval processes.

The trial programme is structured to evaluate:

- CAD-BOM synchronization accuracy and discrepancy detection
- Dependency-aware impact propagation workflows
- Multi-society compliance validation accuracy and traceability
- Drawing synchronization and revision management workflows
- Classification submission workflow support
- Dashboard usability and engineering workflow integration
- Role-based access control and audit traceability
- Air-gapped operational deployment readiness

5.2 Trial Phasing

Phase	Purpose	Indicative Activities
Phase A : Controlled Validation Trial	Validate core engineering workflows using representative historical datasets	CAD-BOM validation, compliance checks, dependency propagation verification
Phase B : End-User Acceptance Trial	Validate operational usability and workflow alignment with MDL engineering teams	Dashboard workflows, revision management, approval workflows, discrepancy review
Phase C : Production Readiness Verification	Validate deployment readiness and operational governance controls	Air-gap verification, audit validation, RBAC verification, deployment readiness review

5.3 Representative Validation Scenarios

Scenario	Validation Objective
CAD-BOM Discrepancy Detection	Verify identification of missing, duplicate, or inconsistent engineering records
Engineering Change Impact Analysis	Validate dependency-aware propagation across affected subsystems
Multi-Society Compliance Validation	Confirm detection and traceability of compliance findings
Drawing Revision Synchronisation	Verify propagation of approved revisions into associated engineering drawings
Classification Submission Workflow	Validate generation and traceability of submission-ready engineering packages
RBAC & Approval Workflow Validation	Verify controlled access and approval hierarchy enforcement
Audit Trail Verification	Confirm traceability and integrity of engineering workflow records
Air-Gapped Operational Verification	Validate complete offline operational capability without external connectivity

5.4 Acceptance Criteria

Validation Area	Indicative Acceptance Objective
CAD-BOM Synchronisation Accuracy	Target validation against representative engineering datasets
Compliance Validation Reliability	Identification of critical compliance findings within validated rule scope
Impact Propagation Coverage	Alignment with SME-reviewed engineering dependency expectations
Drawing Synchronisation Accuracy	Correct revision propagation and traceability across linked drawings
Dashboard Usability	Positive operational usability feedback from participating engineering personnel
Audit & Traceability Verification	Validation of workflow traceability and approval logging

RBAC Enforcement	Verification of role-based operational restrictions and approval workflows
Air-Gapped Deployment Verification	No dependency on external connectivity during operational execution

5.5 User Acceptance & Operational Sign-Off

User Acceptance Testing (UAT) activities will be conducted jointly with authorised MDL engineering, programme management, quality assurance, and IT personnel within the designated deployment environment. The objective of the UAT phase is to validate that AURORA operates reliably within representative naval engineering workflows while meeting operational expectations for traceability, usability, compliance assurance, and deployment readiness.

Throughout the trial process, all validation activities, engineering observations, workflow outcomes, usability feedback, discrepancy findings, and corrective actions will be formally documented and reviewed through structured acceptance and governance procedures.

Final operational acceptance and deployment readiness assessment will be subject to:

- Successful execution and completion of all agreed validation and workflow scenarios
- Verification of CAD-BOM synchronization accuracy, compliance traceability, and dependency-aware impact analysis workflows
- Validation of drawing synchronization, revision management, and classification submission support workflows
- Confirmation of role-based access control, audit integrity, and engineering approval governance mechanisms
- Resolution, mitigation, or formally documented acceptance of identified critical defects and operational observations
- Confirmation of stable operation within the designated air-gapped deployment environment
- Positive usability and workflow suitability assessment from participating MDL engineering personnel
- Verification of deployment readiness, operational maintainability, and traceability requirements
- Final review, acceptance recommendation, and sign-off by authorised MDL stakeholders and programme authorities

Operational deployment progression will occur only after completion of agreed acceptance activities and formal approval through MDL's designated engineering and programme governance processes.

6. Applicable QA & Military-Grade Standards

6.1 Quality Assurance & Engineering Governance Framework

AURORA's development and deployment follows a structured quality assurance framework aligned with defence software engineering requirements:

QA Domain	Assurance Approach
Development Lifecycle Governance	Structured phased development with milestone-based review and gated progression between implementation phases
Configuration & Version Management	Version-controlled engineering artefacts, deployment packages, and rule catalogs with traceable revision management
Rule Validation Governance	Subject Matter Expert (SME) review and validation workflows for all operational rule catalogs prior to activation
Audit & Traceability Integrity	Immutable and traceable logging of engineering actions, approvals, validation outputs, and workflow events
Engineering Change Control	Controlled change-management workflows with impact assessment prior to operational deployment
Documentation Governance	Maintenance of architecture documentation, operational manuals, interface specifications, and deployment records throughout lifecycle phases
Deployment Validation	Structured integration, UAT, and operational verification prior to production rollout

6.2 Engineering & Classification Standards Integrated into AURORA Rule Framework

The following engineering and classification standards are proposed as reference frameworks for the AURORA Multi-Society Compliance Engine and associated validation workflows. These standards are intended to be consolidated into structured digital rule catalogs for deterministic compliance evaluation and engineering traceability.

Standard / Framework	Application within AURORA
Indian Register of Shipping (IRS) Rules for Construction and Classification of Ships	Primary reference framework for naval and maritime compliance validation workflows
IACS Unified Requirements	Baseline engineering and classification requirements applicable across multiple society evaluations

Lloyd's Register (LR) Rules and Regulations for Ships	Multi-society compliance evaluation and rule conflict analysis
DNV Rules for Classification	Engineering rule validation and deterministic compliance evaluation
Bureau Veritas (BV) Rules for Steel Ships	Secondary classification framework for multi-society validation
American Bureau of Shipping (ABS) Rules for Building and Classing Marine Vessels	Secondary classification reference framework
ISO 10303 (STEP)	Neutral CAD and engineering data exchange interoperability
IEC 60092 Series	Marine electrical installation validation references
IEC 61084-1	Cable management and routing validation references

The above engineering and classification frameworks are intended to be incorporated into AURORA's structured compliance rule catalogs for deterministic validation, traceability, and engineering review workflows. Final operational rule scope, coverage depth, and society mappings will be jointly aligned with MDL engineering requirements during implementation phases.

6.3 Cybersecurity, Government & Defence Assurance Considerations

AURORA is designed with consideration for applicable government, cybersecurity, and defence-oriented deployment practices relevant to secure on-premise engineering systems operating within restricted operational environments.

Framework / Guideline	Relevance to AURORA
STQC (Standardisation Testing and Quality Certification) Guidelines	Reference consideration for secure government software deployment and quality assurance practices
CERT-In Security Guidelines	Cybersecurity, logging, incident response, and secure deployment considerations
National Cyber Security Policy (India)	Cybersecurity and operational security considerations for restricted deployment environments
ISO/IEC 27001	Information security management principles for operational infrastructure and data protection
NITI Aayog Responsible AI Principles	Reference framework for explainable, accountable, and human-supervised AI-assisted engineering workflows

MeitY Responsible AI & Trusted AI Guidance	Consideration for transparency, traceability, and controlled intelligent decision-support functions
IndiaAI Mission (Government of India)	Strategic alignment with secure and scalable indigenous AI-enabled infrastructure initiatives
Zero-Trust Security Principles	Segmented authentication and controlled inter-service trust relationships
Role-Based Access Control (RBAC)	Controlled engineering workflow access and approval governance
Air-Gapped Operational Deployment Practices	Secure isolated deployment architecture with no dependency on continuous external connectivity
Digital Personal Data Protection (DPDP) Act, 2023	Consideration for controlled handling of operational user and audit-related data within on-premise environments

6.4 Software Quality & Validation Metrics

The following indicative software quality and validation metrics are proposed for monitoring throughout AURORA's development and deployment lifecycle:

- Target unit and integration test coverage across core engineering modules
- Structured integration validation prior to pilot deployment activities
- Controlled defect tracking and severity-based resolution workflows
- Validation fixtures for operational rule catalogs and compliance workflows
- Documentation completeness for operational interfaces, workflows, and deployment procedures
- Traceable regression validation following major engineering or rule-catalog updates
- User Acceptance Testing (UAT) and operational verification prior to production rollout

7. Other Aspects

7.1 Challenges & Mitigation Strategy

Challenge	Risk	Mitigation
CAD Format Fragmentation	Shipyard may use custom or legacy formats not natively supported by OCC	OCC supports 8+ formats; Phase 0 format assessment; LibreCAD/SolidWorks API bridges as fallback for unsupported formats
Graph Construction Quality	Incomplete BOM-to-dependency mappings produce false negatives in impact analysis	Phase 0 constraint catalogue; SME-reviewed seed templates; ≥ 95% coverage gate before go-live; isolated node flagging
Classification Rule Freshness	Naval classification rules updated periodically; stale rules cause false compliance passes	Structured rule catalog with effective_date fields; quarterly update process in FOM; IRS/DNV/BV rule feeds monitored; PKI-signed offline update packages
Air-Gap Software Updates	On-premise deployment complicates model retraining and software updates	Containerised deployment with offline update packages; retraining pipeline in handover documentation; all artefacts version-controlled in MinIO
Rule Digitisation Complexity	Classification rules are dense semi-structured documents; incomplete extraction causes false compliance passes	Template-based parsing with SME validation loops; phased 30% critical rules in Phase 1; SME sign-off mandatory before rule activation; known-good/known-bad test fixtures
PLM/ERP Integration Complexity	MDL's PLM/ERP environment may differ from standard adapter assumptions	Phase 0 integration assessment collects actual interface specifications; adapter customisation scope agreed before Phase 3 integration work

7.2 Scalability & Future Extensibility

AURORA is designed using a modular multi-layer architecture that supports phased deployment, incremental capability expansion, and long-term scalability across evolving naval engineering

programmes. Individual platform components and analytical engines are designed to operate as independently deployable services, enabling future enhancements without disruption to existing operational workflows.

The platform architecture supports:

- Scalable dependency analysis across large vessel assemblies and interconnected engineering systems
- Incremental expansion of rule catalogs, engineering domains, and validation workflows
- Integration with additional CAD, PLM, ERP, simulation, and programme management ecosystems
- Expansion to additional classification societies, engineering frameworks, or defence-specific validation requirements
- Future enhancement of engineering analytics, optimisation workflows, and advanced design-assistance capabilities
- Flexible deployment across pilot, single-site, and enterprise-scale operational environments

The architecture additionally supports future operational enhancements including expanded engineering validation workflows, advanced geometric analysis capabilities, enhanced cybersecurity hardening, and deeper integration with enterprise engineering ecosystems operational within MDL.

7.3 Human-in-the-Loop Assurance

AURORA is designed with mandatory human oversight integrated throughout all engineering validation, approval, and operational decision workflows. The platform functions as an engineering decision-support environment, while all final engineering judgement, approval authority, and operational release decisions remain under authorised MDL personnel.

Human review and approval checkpoints are integrated across:

- Compliance validation and engineering rule assessment workflows
- CAD-BOM discrepancy review and reconciliation activities
- Engineering optimisation and trade-off analysis outputs
- Drawing revision and production-release processes
- Classification submission and engineering approval workflows
- Operational overrides, exception handling, and workflow escalation procedures

All engineering overrides, approvals, and workflow actions are recorded within the platform's audit and traceability framework together with associated user attribution, timestamps, and operational justification records.

Final engineering authority, certification, and manufacturing release decisions remain entirely with MDL's designated engineering and programme management personnel at all times.

7.4 Institutional Knowledge Retention & Programme Continuity

Beyond its analytical and validation capabilities, AURORA is designed to function as a long-term institutional engineering knowledge-retention platform supporting programme continuity across naval vessel lifecycles.

The platform systematically captures engineering dependencies, compliance relationships, revision history, validation outcomes, and historical change propagation patterns into a structured and traceable operational knowledge framework. This enables preservation of critical engineering context that would otherwise remain distributed across individual domain expertise, isolated documentation systems, and programme-specific operational memory.

The knowledge-retention framework supports:

- Persistent mapping of ship-system interdependencies and engineering relationships
- Traceable historical evolution of engineering and classification rule interpretations
- Searchable reference history of engineering modifications and downstream impacts
- Long-term continuity of audit trails, approval workflows, and revision history
- Improved onboarding and knowledge transfer across engineering teams and vessel programmes
- Reduced dependency on tacit programme knowledge and isolated institutional memory

This capability directly supports long-duration naval programme continuity, engineering traceability, and organisational resilience across evolving vessel classes and engineering lifecycle phases.

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